

The Exciton Model for Molecular Materials: Past, Present and Future?

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In assemblies of identical molecules or chromophores, electronic excitations can be described as excitons, bound electron-hole pairs that can move from site to site as a pair in a coherent manner. The understanding of excitons is crucial when trying to engineer favorable photophysical properties through structur-

ing organic molecular matter. In recent decades, limitations of the concept of an exciton have become clear. The exciton can hybridize with phonon and photons. To clarify these issues, the exciton is discussed within the broader context of the gauge properties of the electromagnetic force.

1. A Brief Introduction to Excitons

Exciton refers to a bound pair of an electron and an electron-vacancy or hole, that can move in a coherent, particle-like manner in an assembly of molecules, atoms, or ions. The concept was first proposed by Frenkel in 1931^[1] and has since found widespread application in the description of excited states in multichromophoric molecules, biopolymers, molecular aggregates, and (molecular) crystals.^[2]

In this contribution, we restrict ourselves to organic, molecular materials. The paper has been organized as follows. In the first introductory part, the focus is on excitons in molecules and molecular systems that are small compared to the wavelength of light. Next, we look at the influence of molecular vibrations. In the third part we will change length scale and look at bulk materials. Finally, the last section involves bulk materials with two different molecular components with application in photovoltaic energy conversion.

For a start, let us first imagine an assembly of identical chromophoric molecules at close distance, yet with an intermolecular separation that is large enough for overlap of molecular orbitals between neighboring molecules to be negligible. The bacteriochlorophyll molecules in the photosynthetic light harvesting complex provide an example.

Here each molecule has a local excited state consisting primarily of an electron excited from HOMO to LUMO. The electron in the LUMO and the vacancy in the HOMO form an electron-hole pair localized on the same molecule and bound by the coulombic attraction between them. Yet within the molecular assembly such an electron hole pair will not remain

stationary but will delocalize over neighboring molecules even though the orbital overlap is vanishingly small. The reason for this delocalization is through-space interaction between the molecules via the electromagnetic field. This interaction can most easily be understood by recalling the semiclassical Lorentz oscillator model. Here an harmonically bound charge vibrates at a particular frequency resulting in emission or absorption of light. When placed at short distance, two Lorentz oscillators of the same resonance frequency will couple via the electromagnetic fields they generate. The system of the two coupled oscillators will exhibit two new resonance frequencies, one corresponding to the in-phase movement and the other to out-of-phase motion, see Figure 1. The difference in frequency and the ordering of the in- and out- of phase combination depends on the relative orientation of the oscillating dipoles in space. It follows quite naturally that for two oscillating dipoles in an in-line, coaxial geometry, the in-phase combination of the dipoles will have the lowest frequency. The frequency splitting is proportional to μ^2/R^3 with R the distance between the centers of the dipoles and μ the electric transition dipole moment.

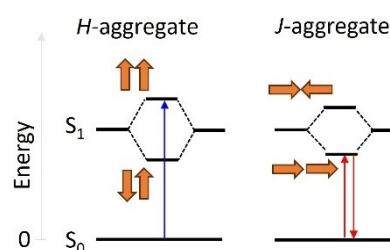


Figure 1. Schematic drawing of two interacting transition dipoles. For the cofacial geometry on the right, the out-of-phase combination of the oscillating dipoles has the lowest energy or frequency. Yet this combination has no net oscillating dipole moment and the transition from the ground state is forbidden. All the dipole strength has accumulated in the transition to the higher exciton level. Thus, the transition is hypsochromically shifted compared to the monomer, and the aggregate is referred to as *H*. In contrast, for the in-line, coaxial geometry on the right, the in-phase combination has the lowest energy and the transition from the ground state to this lower level also accumulates all oscillator strength. According to Kasha's rule, this aggregate should be strongly fluorescent and is named *J* after one of its discoverers (Jelly).

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For parallel dipoles in cofacial orientation, the out-of-phase combination will have the lowest frequency and the splitting is twice as large as for the in-line case.

Importantly, if we now look at the coupled dipoles from far away, we note that only the in-phase combination of the oscillating dipoles will contribute to radiation in the far field limit. In the out-of-phase combination, the contributions from the two opposing dipoles cancel each other at large distance. The red shift of the absorption maximum upon aggregation seen for so-called *J*-aggregates is then explained in terms of a packing of the molecules such that the transition dipoles are oriented mainly in an in-line, coaxial fashion.

One of the charms of the exciton model is that it often caters simple examples of complex physical phenomena. To illustrate this, consider for a *J*-type dimer the magnitude of the total transition dipole moment for the transition between the ground state and the lowest exciton state involving the in-phase combination of the individual transition dipoles. Adding up the dipoles vectorially and taking into account the normalization of the wavefunction, one finds that the total transition dipole moment is a factor of $\sqrt{2}$ larger than the dipole moment of an isolated chromophore. The dipole strength $|\mu_{\text{dimer}}|^2$ is conserved and is $2 |\mu_{\text{monomer}}|^2$ as the transition to the upper exciton level is forbidden. For a *J*-aggregate the transition dipole moment of the lowest allowed transition scales as $\sqrt{N}$ with N the number of molecules over which the excitation is shared.^[3,4] The lowest exciton state is therefore said to be *superradiant*, a term first coined by Dicke.^[5] Note however that in the case of an aggregate, the superradiance already manifests itself for a single excitation. The original proposal by Dicke is more general, in the sense that the superradiance may also be brought about by phase locking several excitations generated via e.g., multiple independent photon absorption events.

The semiclassical Lorentz model can be linked in a straightforward manner to a simple molecular quantum chemical description of the two chromophoric sites involved. Quantum chemical calculations yield an electronic wavefunction for ground and excited states. Multiplying the ground state wavefunction with an excited state wavefunction yields, formally, a probability distribution for the electron. This distribution is known as the transition charge density. This charge density oscillates at the frequency associated with the transition and replaces the oscillating charge in the primitive Lorentz oscillator model. The transition charge density has a

oscillating electric dipole moment, and the magnitude of this lowest order moment largely controls the effectiveness of electromagnetic waves in inducing the transition.

The coupled oscillator model can easily be extended to more complex geometries. Imagine a staircase-like, helical arrangement of two oscillating dipoles, see Figure 2. For this chiral arrangement we can then investigate the difference in interaction with left and right circularly polarized light and try to predict the circular dichroism.

If the two chromophoric units are combined in a right-handed helical geometry, the in-phase combination of the oscillating dipoles will have the higher frequency. The spatial arrangement of the dipoles in this combination is now such that it interacts most strongly to right circularly polarized (RCP) light; the helicity of the local momentary electric dipoles in the scaffold has the same helicity as the electric field in right circularly polarized light. Hence RCP will be more strongly absorbed by the molecular assembly in the high frequency absorption band. The out-of-phase combination of the dipoles results in an absorption at lower frequency compared to the

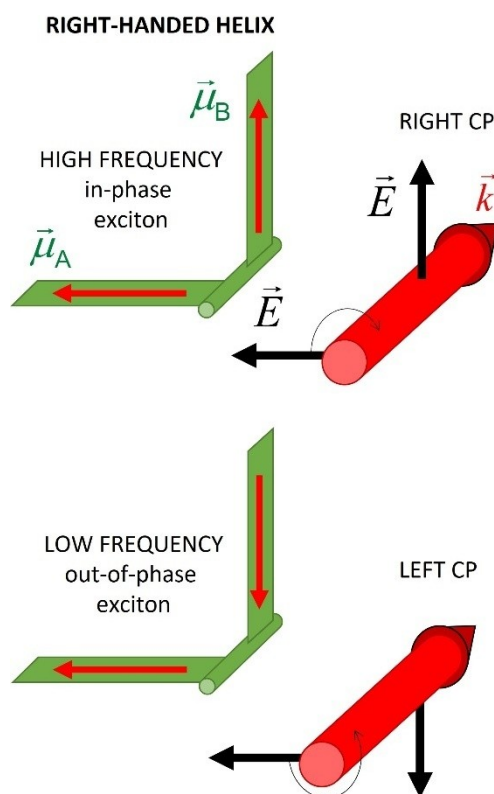


Figure 2. Schematic drawing of a bichromophoric molecule with axial chirality, e.g. binaphthyl. The two chromophoric groups A and B have transition dipole moments μ_A, μ_B that interact. Due to electrostatic repulsion between the like-charged ends of the dipoles, the in-phase combination of oscillating dipoles resonates at higher frequency, the out-of-phase combination at lower. (Upper part). The dipoles in the in-phase combination have a right-handed helicity, the same helicity as the electric field of a right circularly polarized (RCP) photon with wavevector k . RCP photons are preferentially absorbed in the transition from ground state to the high frequency exciton state. (Lower part). For the transition from the ground state to the lower exciton level, the out-of-phase combination of transition dipoles has left-handed helicity and LCP photons are preferentially absorbed.



During his study of chemistry, Stefan Meskers became interested in molecular chirality and the possibility to study it using circularly polarized light. In recent years focus has shifted from chiral molecules and aggregates in solution to molecular solids and their functional properties. The fascination with polarized light has remained.

isolated chromophore. Here the helicity of the dipoles is the same as the helicity of left circularly polarized light. Hence on the low frequency side, one expects absorption of predominantly left circularly polarized light and thus positive circular dichroism. The coupled oscillator model, a special guise of the more general exciton model, provides an intuitive, simple, and yet powerful tool to relate the frequency, intensity and polarization of absorption and emission bands of multichromophoric molecules and supramolecular assemblies to their internal geometrical structure.^[2]

Chirality is a *non-local* geometrical property of an object. One cannot decide whether some structure is chiral by looking only at a part. Even if some part is indeed chiral, it could still be that the other part contains the mirror image of the first part, rendering the total structure achiral. The differential absorption of left and right circularly polarized light by chiral molecules as considered above thus provides an example of a *non-local* response of molecules to electromagnetic radiation. Such a non-local response can occur if the ratio of the typical length a of the molecule and the wavelength of light λ takes on a finite value and can no longer be approximated to zero: $a/\lambda > 0$. The non-local response can be described in various ways. In the above example we have used two local electric dipoles at different positions. Alternatively, one can also introduce higher moments of the transition charge density and the magnetic transition dipole moment is then the lowest term which allows for non-locality.

The development of exciton theory took place largely in the same time window as the genesis of quantum field theory and the standard model unifying the electromagnetic and nuclear forces. The latter have brought us a deeper understating of the electromagnetic interaction. The electromagnetic force differs from the nuclear forces through its unique gauge symmetry. The wavefunction for an electron can be multiplied by an arbitrary phase factor $e^{i\alpha}$ (with e Euler's and i the imaginary number) without altering its probability distributions: the $U(1)$ gauge symmetry of the electromagnetism. There is no absolute way for us to assign the phase of a wavefunction and in our description it should always be possible to change this phase (gauge transformation). From this indeterminacy or symmetry, the Maxwell equations for electromagnetism can be derived.^[6] Also to Lorentz oscillators we cannot assign an absolute phase, we can only determine relative phase differences using electromagnetic radiation. Gauge and gauge transformations turn out to be some of the most fundamental aspects of matter. As we will see in this review, the role and utility of gauge and gauge transformations for understanding spectroscopic transitions and excitons is now slowly becoming clear. By also considering gauge, many of the difficulties we face in exciton theory can be put in a broader perspective and hopefully resolved.

2. Excitons in Helical Polymers

The exciton model has been successful in explaining the electronic circular dichroism of proteins.^[7] The peptide backbone of proteins can be viewed as a collection of amide -HN-

(C=O)- chromophores separated by saturated carbon atoms. The amide chromophore has optical transitions in the deep UV. Interactions between the transition charge densities of nearby amide groups depend on the folding of the peptide chain, i.e., on the secondary structure of the protein. The different folds of the chain (e.g., α -helix, β -sheet, etc.) each give rise to a characteristic circular dichroism spectrum. In practice, signatures of α -helices are easily recognized, allowing the physiological role of this fold in particular proteins to be studied.^[8]

The exciton model has also been applied to intense infrared absorption bands associated with the C=O stretching motion of the amide group (amide I band, near 1650 cm^{-1}). Here β -sheet is the type of secondary structure that is most easily recognized through a characteristic split of the amide I band into a main band near 1630 cm^{-1} and a side band near 1690 cm^{-1} .^[9]

Developments in supramolecular^[10,11] and polymer chemistry^[12] have resulted in synthetic helical chains that allow us to test the exciton model in unprecedented detail. In particular, the role of molecular vibrations in the electronic transitions and electronic circular dichroism has come into focus. We discuss here an oligomeric model compound^[13f] of the newly developed square helical conjugated ladder polymers,^[13] see Figures 3,4. This molecule contains two identical chromophoric groups, namely planarized, ladder type tetra *para*-phenylene. The two *p*-phenylene fragments are held in a chiral arrangement by a bridging spiro carbon atom. Both enantiomeric forms are available.

The exciton model shown in Figure 2 applies to these compounds. Indeed, spectroscopic data show circular dichroism and circular polarization of luminescence from the lowest exciton level consistent with the handedness of the molecules. The fluorescence spectrum shows vibronic progressions based on totally symmetric vibrational modes, indicating that in its lowest electronic excited state, the isolated chromophore adopts a nuclear geometry that is slightly different from the ground state structure. The excited state has more quinoidal character. Looking at the degree of circular polarization of luminescence $g_{\text{lum}} = 2(I_L - I_R)/(I_L + I_R)$, we find that the degree of

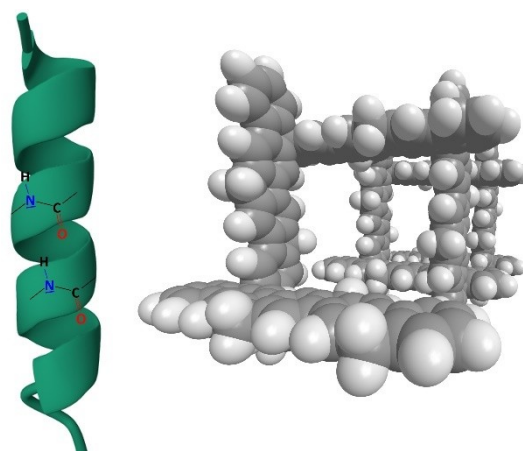


Figure 3. (left) α -helical peptide segment showing also the chemical structure of the HNC=O amide chromophores. (right) Square helical poly(ladder *para*-pentaphenylene) polymer.^[13f]

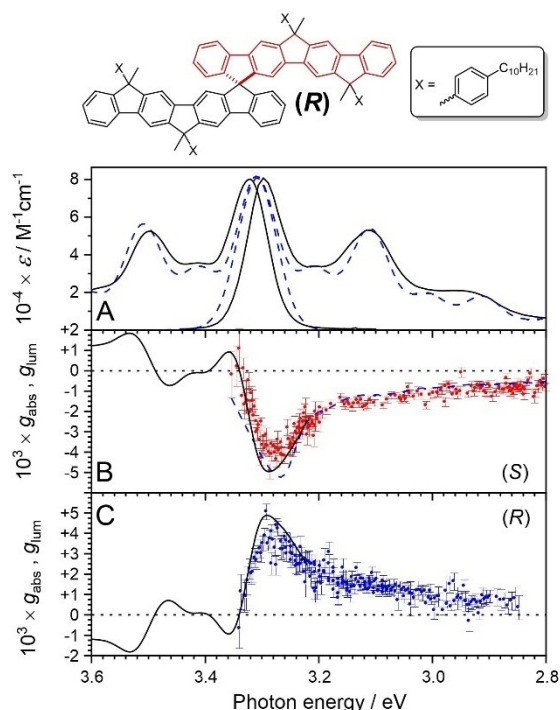


Figure 4. (top) Structure of the dimeric model compound with *R* absolute chirality at the spiro carbon atom. A. Molar absorption and Fluorescence spectrum in THF solution at room temperature. B Degree of circular polarization in absorption ($g_{\text{abs}} = 2(A_L - A_R)/(A_L + A_R)$) and in fluorescence ($g_{\text{lum}} = 2(I_L - I_R)/(I_L + I_R)$) of the (*S*) enantiomer. C. (*R*)-enantiomer. The dashed blue lines represent a quantum chemical model incorporating coupling between exciton and vibrations.^[13f]

circular polarization is relatively high near the onset of the fluorescence but strongly suppressed for the higher vibronic bands at longer wavelength. The lower degree of circular polarization indicates that vibrational motion in the excited state can cause a temporary or dynamical localization of the electronic excitation on one of the chromophores. Recall that the individual chromophores are achiral and that the circular polarization arises from the *coupled* oscillations of the two groups held in a chiral arrangement. The change in degree of circular polarization across the emission band shows directly that the excited state *cannot* be expressed as a single product of an electronic and nuclear (vibrational) wavefunction as would be allowed by the crude Born-Oppenheimer approximation.^[14] Instead the motion of the exciton and vibrations must be coupled. In the limit of weak electronic coupling between the transition dipole moments of the dimers, a relatively simple expression for the excited state wavefunction can be given that still captures some of the complexities of the coupled motion of nuclei and electrons. In this picture we describe the lowest exciton state as combination of two basis states, see Figure 5. In one of the basis states, denoted as $|1_0 0_0\rangle$, chromophore A is in the first excited state (1) and the lowest vibrational level of the relaxed excited state geometry (δ) while chromophore B is in the lowest vibrational level of its electronic ground state 0. If we combine this basis state with the complimentary basis state $|0_0 1_0\rangle$ having the excitation on the B chromophore, we get for

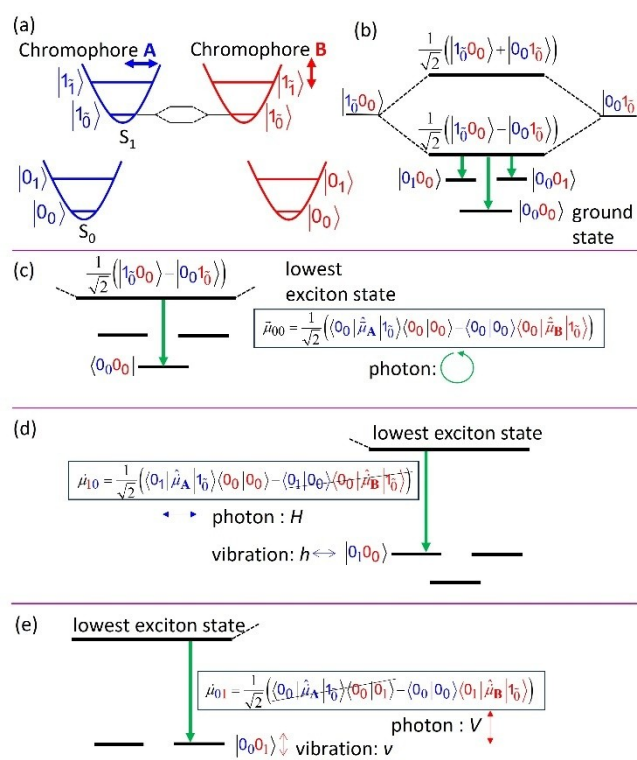


Figure 5. Schematic representation of the exciton model for the bichromophoric molecule in Figure 4 in the limit of weak electronic coupling between the chromophores. (a) ground and excited state potential energy surfaces for the two chromophores. In the relaxed excited state, the vibrational wavefunctions differ from those in the ground state and are indicated by a tilde. The transition dipoles of the two chromophores have different orientation, horizontal for A, and vertical for B. (b) Zoom-in on the lower exciton states. From the lowest exciton states three transitions are indicated, to the ground state, to the vibrationally excited ground state with one vibrational quantum on chromophore A and to the ground state with one vibrational quantum on B. (c) The transition from the lowest exciton state to the true groundstate has contributions from both the transition dipole moments on A and B and is partially circularly polarized. (d) The transition to vibrationally excited ground state on A, is caused solely by the transition dipole on chromophore A and is polarized linearly in the horizontal direction, *H*. (e) the transition to the vibrationally excited ground state on B builds upon the transition dipole moment on B and is polarized vertically *V*.

the total wavefunction of the lowest exciton level:

$$\psi = \frac{1}{\sqrt{2}}(|1_0 0_0\rangle - |0_0 1_0\rangle)$$

This wavefunction is no longer a single product of a nuclear and electronic function. The coupling of electronic and vibrational motion in this state is apparent from the fact that the electronic excitation (1) and the distortion of the nuclear geometry (δ) always occur on the same chromophore. If we now compute the transition dipole moments for the transition from this lowest exciton level to the lowest vibrational level of the ground state ($|0_0 0_0\rangle$) we find that both chromophores contribute equally and given their spatial arrangement, the luminescence should have some circular polarization. If we then consider the transition to the first vibrationally excited level of the electronic ground state with the ground state vibration located on chromophore B ($|0_0 0_1\rangle$), we find that only

chromophore B, i.e. the chromophore that retains the molecular vibration in the final state, can contribute. In addition, there will be an overlapping, corresponding transition in which only the transition dipole moment of chromophore A participates ending in the vibrationally excited ground state with the vibration located on A. Thus, on average, the second vibronic band of the fluorescence ending in a vibrationally excited level of the ground state will have a vanishingly small circular polarization in the weak coupling limit considered here. All the circular polarization should be concentrated in the first vibronic band, often referred to as 0–0 transition.

This seemingly isolated, innocuous spectroscopic detail involving the coupled motion of electrons and nuclei may in fact have wider implications in the field of quantum information. To see this, we consider the final state for the weakly coupled dimer as discussed above and look at the linear polarization of the photon emitted via the 0–1 transition ending in a vibrationally excited level of the ground state. Each individual photon emitted via this transition will be polarized either parallel to the transition dipole moment of chromophore A or parallel to chromophore B. For the particular molecule under study here, we could assign the polarizations horizontal *H* and vertical *V* to these two possibilities. Interestingly if the transition proceeds via the transition dipole moment of chromophore A with the emitted photon being *H* polarized, we also know that the nuclear vibration must have ended up on chromophore A. Assuming that the vibration is along the long axis of the chromophore, we can label the vibration as *v*. Now the most general wavefunction for the final state ϕ before doing any measurement on the system is then:

$$\phi = \frac{1}{\sqrt{2}}(|Vv\rangle - |Hh\rangle)$$

In general, such a state function incorporating multiple products of wavefunctions indicates entanglement. In this particular case, the photon polarization is entangled with the vibration. If one would measure the polarization of the emitted photon and find say *V*, the wavefunction would collapse immediately and vibration should be *v*. This intriguing prediction of entanglement between a photon and a molecular vibration still needs to be tested experimentally.

The Born-Oppenheimer separability of nuclear and electronic motion is problematic when two electronic states are close in energy. Yet the crossing of electronic states in so-called conical intersections is crucial in understanding (photochemical) reactivity.^[15] In essence the Born-Oppenheimer involves ignoring the dependence of the motion of electrons on the speed of the atomic nuclei. This may be difficult to visualize, especially given the way the potential energy for the electrons as set up by the positively charged atomic nuclei is usually represented in the Hamiltonian. Importantly, there is no unique way to represent potentials and there are in fact many equivalent ways to formulate potentials and Hamiltonians.^[16] Using gauge transformations these can be transformed into one another. In quantum chemistry it is customary to use the so-called Coulomb gauge. A peculiar feature of this particular gauge is

that the scalar potential Φ is unretarded. Φ enters the Hamiltonian for electrons directly as electric potential *V*. The potential due to a particular nucleus then depends only on the instantaneous position of that nucleus. So, when writing down a Hamiltonian for the electrons, there appears to be no dependence on how fast the nuclei move because only their instantaneous position occurs. The dependence of the electronic motion on momenta (speed) of the nuclei is hidden. An alternative way to formulate the Hamiltonian is to involve the Lorentz gauge. Here all interactions are retarded and interactions between nuclei and electrons occur via the electromagnetic field through exchange of virtual photons. Unfortunately, quantum chemical calculations involving Lorentz gauge are highly impractical and no procedures or routines are available. Still the formulation of the Hamiltonian using Lorentz gauge makes it much easier to visualize the exchange of energy between electronics and nuclei and the dependence on the speed of the nuclei. Just imagine two electronic states separated by a small energy corresponding to the frequency of nuclear motion. The electrons and nuclei can now exchange energy via emission and absorption of virtual photons. Emission of such a virtual photon by nuclei is only possible if the nuclei move. Hence the dependence of electrons moving from one potential energy surface to another on the speed of nuclei becomes obvious.

The close connection between gauge of the electrodynamic potentials and the Born-Oppenheimer approximation has been noted before.^[17] Interestingly, even for electronic ground states of closed shell molecules for which the Born-Oppenheimer approximation is usually assumed to be highly reliable, the non-adiabatic coupling of electronic and nuclear motion can be an issue. It is well known that in the calculation of intensities and circular dichroism for infrared transitions, application of the Born-Oppenheimer approximation leads to different outcomes depending on which gauge (dipole or velocity) is adopted for the potential.^[18] Also calculations of infrared circular dichroism^[19] and time-dependent simulation of electronic optical activity require careful evaluation of corrections to the Born-Oppenheimer approximation.^[20]

In summary, chiroptical properties of helical polymers consisting of non-conjugated chromophoric segments can be explained in terms of excitons. Coupling of the exciton with nuclear vibrational can lead to dynamical localization of the exciton.^[21] This involves breakdown of the Born-Oppenheimer separability of electronic and nuclear motion. For the broadest possible perspective on these complex coupled motions, also the gauge used in formulation the Hamiltonian needs to be taken into consideration.

So far, most studies of the interaction of excitons with phonons have focused on quasi one-dimensional systems such as polymers or fiber-like aggregates.^[22] In one-dimensional systems, localization is often most severe as there is essentially no possibility for quasi particles to go around deformations. Interestingly for systems with dimensionality higher than one, it has been argued that localization or self-trapping can be less strict. At least in adiabatic theory, self-trapped and 'free' quasiparticles may even coexist with an energy barrier separat-

ing these two species.^[23] Arguably the strongest experimental support for a barrier separating free and trapped excitons comes from alkali halide crystals^[24] where essentially an optical excitation can convert reversibly into a lattice defect via thermal activation. However also support for the contrary view, i.e., a barrierless transitions between free and trapped excitons, has been put forward,^[25] underlining the complicated but fascinating nature of the electron-phonon problem. In the next section we first look at experimental work on excitons in higher dimensional systems and then we try to put the problem in a broader view involving once again gauge.

3. Excitons in Systems with Dimension Higher than One

Cyanine dyes, known for their ability to form fiber-like aggregates in solution, can also form two-dimensional layers on surfaces.^[26] These packed layers display optical characteristics similar to the quasi-one-dimensional *J*-aggregates in solution: narrow absorption bands and efficient fluorescence. A number of other dye molecules are also known to form 2D *J*-aggregates.^[27,28] Spectroscopic properties consistent with *H*-type aggregation in 2D are also well attested in the scientific literature.^[29]

With regards to three dimensional, bulk materials we first note that organic molecular luminescent materials have become of major importance in electrically driven displays in recent decades. Highly efficient molecular light emitters are available. Yet, although signatures of *J*- and *H*-aggregation in bulk solids have been observed^[30] and evidence of super-radiance in solids has been found,^[31] the hope of finding materials showing superradiance of a type and strength similar to pseudoisocyanine *J*-aggregates in solution has not materialized.

A detailed study of a layer-by-layer deposition of 2D *J*-aggregates to form a 3D material, confirmed strong absorption of light but also showed a gradual broadening of the *J*-band as the number of stacked 2D layers was increased.^[32] A number of studies have focused on the optical properties of bulk crystals of pseudo-isocyanine dye.^[33] The reflection spectra off various faces do not directly feature the ultranarrow linewidths of *J*-aggregates in frozen solution. No fluorescence from the crystals was reported in these studies. Altogether these observations indicate that in three-dimensional bulk, the notion of excitons requires more careful consideration.

When considering the optical properties of the 3D bulk, it can be argued that the notion of radiative decay takes on a different meaning. For quasi zero-, one- or two-dimensional molecular assemblies one could argue that radiative decay involves coupling of an excited state to a continuum of photon states in the surrounding empty space.^[34] In a perfect, infinite crystal however, an exciton has a specific momentum quantum number *k* because of the translational symmetry and can only couple to electromagnetic waves with the same *k* quantum number. The possibility for irreversible decay of the excitation

via coupling to a continuum of electromagnetic waves seems eliminated. Instead, as proposed by Hopfield, the exciton and the electromagnetic waves with the *k* quantum number should be expected to hybridize resulting in a polariton.^[35,36] The crossover from exciton to polariton behavior has been discussed.^[37] A simplistic view of polaritons in bulk is the following. Suppose that through interactions among molecular transition dipole moments in the solid, absorption and re-emission of a photon become much faster than any other photophysical process in the excited state. Then it no longer makes sense to think about an exciton undergoing e.g., relaxation of the nuclear geometry, because before any such process will be complete the exciton will already have converted. Instead, one will have to acknowledge the quantum mechanical mixed photon-exciton character of the quasiparticle when trying to calculate rates. The hybrid photon-exciton quasiparticle is a polariton.

So while exciton theory as proposed originally by Davydov^[38] provides useful insights into the behavior of weak, low energy transitions in e.g. crystals of the acenes, it also faces a number of issues.

- The sum of the interaction energies between all the transition dipoles in the crystal is at best only conditionally convergent.^[39,40]
- For some exciton states, the dipole strength may be proportional to the number of coherently coupled molecules $\mu^2 \propto N$, as we have seen for instance for the lowest exciton state in *J*-aggregates. With such large dipole strength, the interaction energy between the transition dipole moment and the electromagnetic field becomes substantial and the first order perturbation theory normally applied to describe the transition may no longer apply.
- Optics holds that inside the crystal, light waves will travel at a speed different from *c*. For the momentum of these light waves conflicting expression have been offered, leading to the Abraham-Minkowski dilemma.^[41]
- For calculating the reflection of light off a crystal face, one needs boundary conditions for the electromagnetic field. Inside the crystal there are not only the electromagnetic waves but also the electrical polarization *P*. While the boundary conditions for the electric and magnetic waves can be derived from Maxwell's equations, the additional boundary condition (ABC) to be applied to the polarization is controversial.^[42] If one treats the polarization as a strictly local and linear response, then the polarization *P* will have the same *k* vector as the electric field. In this case with *P* proportional to the local value of *E*, i.e., the dipole approximation, a separate boundary conditions for the polarization is not strictly needed. Yet when leaving the dipole approximation to describe e.g., the non-local, chiroptical response of chiral materials, the problem of the additional boundary condition becomes acute.
- Reflection of light at a flat, dielectric boundary in optics proceeds with inversion of circular polarization.^[43,44] This results from the electric field being a vector inverting sign upon reflection, while the magnetic field is a pseudovector that remains invariant upon reflection. Yet, it is also known

that cholesteric liquid crystals can reflect circularly polarized light with retention of circular polarization. This shows that treating reflection of light as a process involving only a vanishingly thin surface layer of the dielectric must be an approximation.

- Upon reflection of light off a flat, dielectric substrate in an off-normal geometry, the phase difference between *s*- and *p*-polarized light, known in ellipsometry as the Δ -angle, can only assume values of 180° or 0° , except when the substrate is absorbing (or covered by a thin layer).^[45] In practice however for bulk materials, one often observes Δ -values different for 180° or 0° at frequencies where absorption is not expected, e.g. in the subgap spectral region of the material.

When trying to describe the optical properties of 3D bulk solids including reflection, one is actually looking for effective waves inside the materials that travel at a speed different from *c* and have integer spin. From relativistic field theory, such waves are known as massive vector boson fields. These vector bosons have three independent polarization components rather than the two independent polarizations for the photon in vacuum. The additional third polarization in the material results from coupling of the pure electromagnetic waves travelling at speed *c* with the mechanical degree of freedom of charged particles in the material associated with electrical polarization.

The most general way to describe electromagnetic waves inside matter is the four-potential *A* rather than electric, magnetic and polarization fields. Moreover, when formulating a Hamiltonian, one needs potentials anyway. The four-potential can be visualized in the following simple way. If we want to set up an arbitrary electric or magnetic field at a certain point in space, we will need four electrodes, each connected to the source. To each electrode, we would then apply a certain potential to get the desired field. The set of these four potentials may then be referred to as the four-potential. Customarily, the first of the four components, is the electrical or scalar potential Φ . The remaining three component are those of the vector potential $\vec{A}$.

There is some redundancy in this set of four numbers: one of the four electrodes may be used as a ground and its potential will always be assigned “0”. Thus, to be able to recreate the desired field, we need to know which electrode should be used as the ground and then three numbers stating the potential to be applied to the remaining three electrodes. Obviously, the ground can be chosen in many different but equivalent ways. This choice is referred to as gauge freedom. This freedom in gauging the field ultimately relates to the arbitrariness of the phase factor of a charged particle as mentioned in the introduction. Several different gauges are possible. Most commonly used gauges are Lorentz gauge and Coulomb gauge. In the Lorentz gauge, the relation between the four components of the four-potential is

$$\frac{\partial}{\partial t}\Phi = \vec{\nabla} \cdot \vec{A}$$

The Lorentz gauge has the advantage that it is directly compatible with special relativity: for every observer moving at

constant speed, formulas involving the four-potential in Lorentz gauge will be the same and all potential waves including Φ propagate at speed *c* in vacuum.

In Coulomb gauge, one requires the vector potential to be transverse:

$$\vec{\nabla} \cdot \vec{A} = 0$$

This means that for any plane four-potential wave, the component of the vector potential in the direction of motion is set to zero.

In quantum chemistry, Coulomb gauge is assumed. Here the advantage is that radiation effects involving the vector potential $\vec{A}$ can be separated out and treated as a perturbation. The scalar potential Φ is included directly in the Hamiltonian. This greatly simplifies quantum chemical calculations. The scalar potential in Coulomb gauge turns out to be unretarded: in this gauge the potential for a particular electron at a certain moment depends upon the *instantaneous* position of the other electrons and nuclei. This is of course incompatible with special relativity which holds to no physical influences can travel faster than light. Still, it can be shown that when all possible radiative corrections involving the vector potential are performed, the final outcome is consistent with special relativity. The problem in practice is however that the radiative corrections are difficult to calculate, and the perturbations series used in the calculations may not converge. Still, the Coulomb gauge is extremely versatile in quantum chemistry because for small systems like the hydrogen atom, the radiative corrections needed in the energy of the orbitals are extremely small (1 in 10^6) as confirmed experimentally by measurements of the Lamb shift. However, the radiative corrections needed can become much larger for heavy atoms, molecules^[46–49] and mesoscopic resonator circuits.^[50]

The Lorentz gauge has the advantage that knowing the local response of a material in terms of resonance frequency and multipole moments, one can calculate long-range correlation effects in electronic motion via the electromagnetic field. It is well appreciated that London dispersion forces involving induced dipole-dipole interaction via the electromagnetic field are covered by electron-electron correlation^[51,52] and may also be used to introduce correlation effects in density functional theory.^[53]

Now for the excitons in macroscopic molecular crystals, the use of Coulomb gauge becomes problematic. When calculating the energy of the excitons, one neglects radiation effects and just takes the *unretarded* dipole-dipole interaction consistent with Coulomb gauge. A major problem is now that the sum of all unretarded dipole-dipole interactions is only conditionally convergent. Retarded dipole-dipole interactions, as occur in the Lorentz gauge description, have much better convergence properties. So, it seems that for the 3D solids, separating out radiation effects in the quantum chemistry approach relying on Coulomb gauge, turns out to be unfortunate in the sense that the separate calculations of electronic states involving the dipole sums and the radiative correction needed afterwards, yield divergencies that should ultimately cancel each other.

The concept of gauge may also clarify the issue of the additional boundary condition. To explain this, we first need to consider electromagnetic waves in empty space. In absence of charges, all electromagnetic waves travel at the speed of light. If we would try to recreate such a moving electromagnetic field with our system of four electrodes, the electrodes would have to be moving at speed c and would undergo a complete Lorentz contraction in the direction of motion. As a result, we cannot establish an electric field with a component in the direction of motion, see Figure 6. This means that yet another of the four parameters of the four-potential can be eliminated. This is often referred to as additional gauge freedom. For massless boson fields, i.e., boson fields propagating at the speed of light, only two independent polarization states are available. In coulomb gauge, this implies that also the scalar potential can be set to zero. In Lorentz gauge, the component of the vector potential in the direction of motion may still be non-zero, provided that the electric field E in the direction of motion vanishes.

When calculating reflection at the interface between vacuum and a material half space, the additional gauge symmetry on the material side is obviously broken because the effective waves in the material travel at speed c/n with n the refractive index. This breaking of symmetry permeates the whole space in the reflection problem because we need to introduce a uniform gauge at the interface. The boundary conditions for the four-potential in the reflection problem can now be supplemented with an addition condition of uniform gauge.^[54]

The discussion of gauge above also makes clear that components of the radiation field in a Lorentz gauge description can get included in the exciton wavefunctions through the scalar potential in the Coulomb gauge picture. Therefore, it is no surprise that alternative formulations for the momentum of electromagnetic radiation in a solid can be given. Thus, the Abraham-Minkowski controversy should be

viewed in a broader context also including the gauge conditions applied.

In summary, in multichromophoric molecules and one-dimensional aggregates, electronic excitations and radiation can be separated effectively in the conventional quantum chemical treatment relying on Coulomb gauge. In bulk molecular solids, excitations and radiation become intimately coupled. The system of coupled equations describing electron and field motion can be formulated in several ways, and this gauge freedom can help to clarify and resolve inconsistencies. In particular in recent years, experimental support for polaritons rather than excitons as primary excitations in resonant cavities^[55,56] and also in bulk molecular solids has been presented.^[57,58] In addition, the exciton may not only hybridize with a photon but also mix with molecular vibrations.^[59]

4. Excitons in Organic Heterojunctions

One of the most astonishing developments in the field of organic molecular materials is the realization of highly efficient organic solar cells.^[60–63] These cells generally involve a (bulk) heterojunction between an organic molecular material with electron donating and one with electron accepting properties, see Figure 7. The perhaps surprising finding is that excitons, despite being tightly bound electron-hole pairs in the pure donor and acceptor materials, dissociate rapidly at the interface between the two materials into separated electrons and holes with almost unit efficiency. In order to discuss this in more detail, we first need to extend our concept of exciton states. So far, we have only discussed singlet Frenkel excitons, where the electron and the hole have antiparallel spin. Also, triplet excitons exist where the electron and hole have parallel spins.^[64] In neat molecular crystals, these triplet excitons usually have an energy that is considerably below that of the lowest singlet exciton, typically 0.6 eV. This relatively large singlet-triplet gap is consistent with a tightly bound electron-hole pair having a considerable exchange energy.

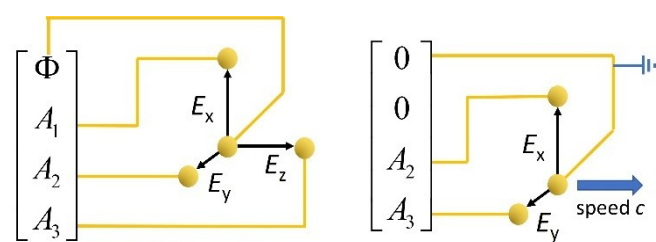


Figure 6. The electromagnetic four-potential. (Left) Four electrodes are needed to recreate an arbitrary electric field (E_x, E_y, E_z) and the electromagnetic four-potential may be interpreted as the set of four voltages to be applied to such a set of electrodes. Of these four potential values, one is redundant because one electrode can be used as a ground. Yet the ground can be chosen in several different ways, indicating gauge freedom. By applying time-dependent voltages, electrodes can be charged temporarily and moving charge can be simulated. In this way also magnetic fields may be recreated. (Right) To recreate the electric field of a light wave moving through vacuum, one could imagine the set of electrodes moving at speed c . Because of the Lorentz contraction, the electrode pointing in the direction of motion will be 'squashed' and not functional anymore. Taking grounding into account also, only two active electrodes are left. Hence light in vacuum has two independent polarization states.

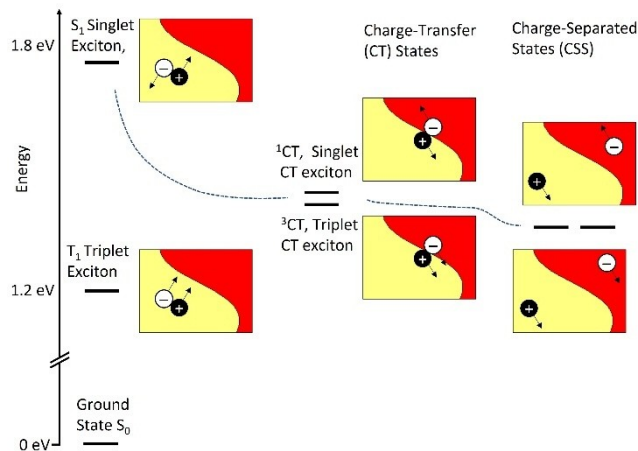


Figure 7. Various types of excitons that exist near the interface between two organic molecular materials, one with electron donating and the other with electron accepting properties (coloured yellow and red respectively).

So far in our discussion of excitons, we have largely neglected exchange of electrons between neighboring molecules. Especially in mixed crystals including molecules with electron donating and electron accepting properties in neighboring lattice positions, this exchange of electrons can be significant. Excitons including a substantial amount of charge transfer are termed charge-transfer (CT) excitons. CT excitons can also be of either singlet or triplet spin multiplicity. Yet because the electron and hole are effectively on different molecules, the exchange interaction and thus the singlet-triplet energy splitting are usually much smaller. Finally, electron and hole may also be separated over several molecular distances, in this case one speaks of charge-separated states (CSS) of the exciton. This is in a sense similar to the weakly bound Wannier excitons encountered in inorganic semiconductors.

The operation of an organic solar cell is currently understood in the following way.^[65,66] Photon absorption in either donor or acceptor material leads to formation of a singlet exciton that diffuses towards the donor-acceptor interface. At the interface electron transfer occurs leading to formation of charge transfer excitons (¹CT, ³CT). These CT excitons may then dissociate further into charge-separated states or recombine to form e.g., a triplet exciton in one of the two materials.

At present it is still not fully clear how the CT excitons can dissociate so efficiently into separated charges at the interface between the two materials despite a substantial exciton binding energy. Numerous experiments have indicated that the degree of segregation between donor and acceptor material is of importance. The rate for charge generation out of excitons k_{CS} may be expressed in terms of its activation barriers:

$$k_{CS} \propto \exp(-\Delta G^{\ddagger}/k_b T)$$

$$\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$$

It should be noted that the effective free energy barrier $\Delta G^{\ddagger}$ in the charge separation process is not only determined by the largely electrostatic activation enthalpy ($\Delta H^{\ddagger}$) but is also influenced by entropic contributions ($\Delta S^{\ddagger}$) related to the number of ways the various states can be realized in the system.^[67] In particular, a large density of charge-separated states relative to the density of exciton states will favor charge separation. Using the kinetic Monte-Carlo simulation method and assuming localized states for both the excitons and charges represented by cubic lattices of localized states in space, one can to some extent account for the densities of states and the activation entropy factor.^[68,69] Yet in general, accurate densities of states require knowledge of the dispersion relations of the excitons that are usually not included in hopping type simulations. An interesting issue here is how coupling of the exciton with phonons and/or photons will influence the density of states. It has already been demonstrated that by tuning the thickness of the active layer and using highly reflective electrodes, the energy of the exciton states can be influenced via light-matter coupling in the resulting optical cavity.^[70] In addition, for some donor-acceptor blends, very large exciton diffusion lengths have been observed, up to 50 nm, suggesting

that exciton delocalization and hybridization with photon modes may be important.^[71] Finally the different response of solar cells made from chiral materials to left and right circular polarized light directly supports a non-local response of the material to incident radiation, prompting the need for a more extended treatment of the excitons.^[72]

3. Summary and Outlook

The concept of excitons is still very much subject of ongoing research into the optical properties of molecular materials. The exciton can couple and mix with molecular vibrations and also with photons. The latter hybridization is especially important in bulk crystals. Efficient computation of the properties of these excitations with mixed electronic, vibrational, and electromagnetic character requires maximal flexibility of the theoretical methods used. To achieve such flexibility, we should make use of gauge transformations as much as possible. An interesting path forward seems to be a coarse graining approach where one applies traditional quantum chemistry in Coulomb gauge to compute excited state energies and multipole transition moments for isolated molecules and/or clusters of limited size. These parameters may then subsequently be used in an effective field theory involving Lorentz gauge.

Conflict of Interests

The authors declare no conflict of interest.

Keywords: Exciton • exciton chirality method • Born-Oppenheimer approximation • polariton • gauge transformation

- [1] J. Frenkel, *Phys. Rev.* **1931**, *37*, 17.
- [2] a) M. Kasha, H. R. Rawls, M. Ashraf El-Bayoumi, *Pure Appl. Chem.* **1965**, *11*, 371; b) N. Harada, K. Nakanishi, *J. Am. Chem. Soc.* **1969**, *91*, 3989; c) N. Harada, K. Nakanishi, *Circular Dichroism Spectroscopy: Exciton coupling in organic chemistry*, University Science Books, Mill Valley CA **1983**; d) T. Brixner, R. Hildner, J. Köhler, C. Lambert, F. Würthner, *Adv. Energy Mater.* **2017**, *7*, 1700236; e) G. Pescitelli, *Chirality*. **2022**, *34*, 333.
- [3] H. Fidder, J. Knoester, D. A. Wiersma, *Chem. Phys. Lett.* **1990**, *171*, 529.
- [4] R. Monshouwer, M. Abrahamsson, F. van Mourik, R. van Grondelle, *J. Phys. Chem. B* **1997**, *101*, 7241.
- [5] R. H. Dicke, *Phys. Rev.* **1954**, *93*, 99.
- [6] a) J. Schwichtenberg, *Non-nonsense Quantum Field Theory*, No-Nonsense Books, Karlsruhe, **2019**; b) L. H. Ryder, *Quantum Field Theory*, 2nd ed., Cambridge University Press, Cambridge, **1996**.
- [7] a) J. G. Kirkwood, *J. Chem. Phys.* **1937**, *5*, 47; b) W. Moffitt, *J. Chem. Phys.* **1956**, *25*, 467; c) I. Tinoco Jr, R. W. Woody, *J. Chem. Phys.* **1960**, *32*, 461; d) J. A. Schellman, *Acc. Chem. Res.* **1968**, *1*, 144.
- [8] P. C. N. Rensen, N. Herijgers, M. H. Netscher, S. C. J. Meskers, M. van Eck, T. J. C. van Berkel, *J. Lipid Res.* **1997**, *38*, 1070.
- [9] a) A. Barth, *BBA-Bioenergetics* **2007**, *1767*, 1073; b) T. Hayashi, S. Mukamel, *J. Phys. Chem. B* **2007**, *111*, 11032; c) Y. Abe, S. Krimm, *Biopolymers* **1972**, *11*, 1817; d) Y. N. Chirgadze, B. V. Shestopalov, S. Y. Vennyaminov, *Biopolymers* **1973**, *12*, 1337.
- [10] T. Aida, E. W. Meijer, *Isr. J. Chem.* **2020**, *60*, 33.
- [11] a) C. Spitz, S. Dähne, A. Ouart, H.-W. Abraham, *J. Phys. Chem. B* **2000**, *104*, 8664; b) S. Kirstein, H. von Berlepsch, C. Böttcher, C. Burger, A. Ouart, G. Reck, S. Dähne, *ChemPhysChem*. **2000**, *1*, 146; c) U. De Rossi, S. Dähne, S. C. J. Meskers, H. P. J. M. Dekkers, *Angew. Chem. Int. Ed.* **1996**, *35*, 760.

- [12] E. Yashima, K. Maeda, H. Iida, Y. Furusho, K. Nagai, *Chem. Rev.* **2009**, *109*, 6102.
- [13] a) Y. Dai, T. J. Katz, *J. Org. Chem.* **1997**, *62*, 1274; b) T. Ikai, T. Yoshida, K. Shinohara, T. Taniguchi, Y. Wada, T. M. Swager, *J. Am. Chem. Soc.* **2019**, *141*, 4696; c) W. Zheng, K. Oki, R. Saha, Y. Hijikata, E. Yashima, T. Ikai, *Angew. Chem. Int. Ed.* **2023**, *62*, e202218297; d) T. Ikai, S. Miyoshi, K. Oki, R. Saha, Y. Hijikata, E. Yashima, *Angew. Chem. Int. Ed.* **2023**, *62*, 20; e) J. Du, H. Li, T. Wu, M. Wang, R. Cheng, D. Wu, Y. Yang, J. Lan, *Chem. Commun.* **2023**, 59, 8957; f) R. Ammenhäuser, P. Klein, E. Schmid, S. Streicher, J. Vogelsang, C. W. Lehmann, J. M. Lupton, S. C. J. Meskers, U. Scherf, *Angew. Chem. Int. Ed.* **2023**, *62*, e202211946.
- [14] W. Siebrand, *Chem. Phys. Lett.* **1977**, *51*, 5.
- [15] D. R. Yarkony, *Acc. Chem. Res.* **1998**, *31*, 511.
- [16] D. P. Craig, T. Thirunamachandran, *Molecular Quantum Electrodynamics*, Academic Press, London, **1984**.
- [17] W. Domcke, D. Yarkony, H. Köppel, *Conical intersections: electronic structure, dynamics & spectroscopy*, World Scientific, River Edge, N. J., **2004**.
- [18] a) N. V. Cohan, H. F. Hameka, *J. Chem. Phys.* **1966**, *45*, 4392–4399; b) C. A. Mead, A. Moscovitz, *Int. J. Quantum Chem.* **1967**, *1*, 243; c) T. B. Freedman, L. A. Nafie, *Chem. Phys. Lett.* **1986**, *126*, 441.
- [19] P. J. Stephens, *J. Phys. Chem.* **1985**, *89*, 748.
- [20] J. Mattiat, S. Lubner, *J. Chem. Theory Comput.* **2022**, *18*, 5513.
- [21] a) M. Deutsch, S. Wirsing, D. Kaiser, R. F. Fink, P. Tegeder, B. J. Engels, *J. Chem. Phys.* **2020**, *153*, 224104; b) E. Titov, T. Kopp, J. Hoche, A. Humeniuk, R. Mitrić, *Phys. Chem. Chem. Phys.* **2022**, *24*, 12136; c) A. Troisi, *Phys. Rev. B* **2010**, *82*, 245202.
- [22] a) F. C. Spano, *Ann. Rev. Phys. Chem.* **2006**, *57*, 217; b) F. C. Spano, Z. Zhao, S. C. J. Meskers, *J. Chem. Phys.* **2004**, *120*, 10594; c) F. C. Spano, S. C. J. Meskers, E. Hennebicq, D. Beljonne, *J. Am. Chem. Soc.* **2007**, *129*, 7044.
- [23] a) D. Emin, T. Holstein, *Phys. Rev. Lett.* **1976**, *36*, 323; b) Y. Toyozawa, Y. Shinozuka, *J. Phys. Soc. Jpn.* **1980**, *48*, 472; c) L.-C. Ku, S. A. Trugman, J. Bonča, *Phys. Rev. B* **2002**, *65*, 1743061.
- [24] K. S. Song, R. T. Williams, *Self-trapped excitons*, Springer, Berlin, **1996**.
- [25] a) A. Lagendijk, H. de Raedt, *Phys. Lett. A* **1985**, *108*, 91; b) A. H. Romero, D. W. Brown, K. Lindenberg, *Phys. Rev. B* **1999**, *60*, 14080.
- [26] a) H. Hada, R. Hanawa, A. Haraguchi, Y. Yonezawa, *J. Phys. Chem.* **1985**, *89*, 560; b) Y. Yonezawa, D. Möbius, H. Kuhn, *J. Soc. Photogr. Sci. Technol. Jpn.* **1985**, *48*, 250; c) C.-H. Tian, G. Zorinians, R. Gronheid, M. van der Auweraer, F. C. De Schryver, *Langmuir* **2003**, *19*, 9831; d) C.-H. Tian, D.-J. Liu, R. Gronheid, M. Van der Auweraer, F. C. De Schryver, *Langmuir* **2004**, *20*, 11569; e) M. Liu, A. Kira, H. Nakahara, *J. Phys. Chem.* **1996**, *100*, 20138; f) V. V. Prokhorov, S. I. Pozin, D. A. Lypenko, O. M. Pereyagina, E. I. Mal'tsev, A. V. Vannikov, *Chem. Phys. Lett.* **2012**, *535*, 94; g) M. Kawasaki, T. Sato, T. Yoshimoto, *Langmuir* **2000**, *16*, 5409; h) R. W. Owens, D. A. Smith, *Langmuir* **2000**, *16*, 562; i) N. Kato, K. Saito, T. Serata, H. Aida, Y. Uesu, *J. Chem. Phys.* **2001**, *115*, 1473; j) M. Orrit, D. Möbius, U. Lehmann, H. Meyer, *J. Chem. Phys.* **1986**, *85*, 4966.
- [27] a) B. Sun, X. Xu, G. Zhou, L. Tao, W. Xinran, Z. Chen, J.-B. Xu, *J. Phys. Chem. A* **2020**, *124*, 7340; b) B. Sun, G. Zhou, Y. Wang, X. Xu, L. Tao, N. Zhao, H. K. Tsang, X. Wang, Z. Chen, J.-B. Xu, *Adv. Opt. Mater.* **2021**, *9*, 2100158.
- [28] E. Da Como, M. A. Loi, M. Murgia, R. Zamboni, M. Muccini, *J. Am. Chem. Soc.* **2006**, *5*, 4277.
- [29] a) E. V. Agina, A. A. Mannanov, A. S. Sizov, O. Vechter, O. V. Borshchev, A. V. Bakirov, M. A. Shcherbina, S. N. Chvalun, V. G. Konstantinov, V. V. Bruevich, O. V. Kozlov, M. S. Pshenichnikov, D. Yu. Parashuk, S. A. Ponomarenko, *ACS Appl. Mater. Interfaces* **2017**, *9*, 18078; b) F. Gholamrezaie, M. Kirkus, S. Mathijssen, D. M. de Leeuw, S. C. J. Meskers, *J. Phys. Chem. A* **2012**, *116*, 7645; c) F. Gholamrezaie, D. M. de Leeuw, S. C. J. Meskers, *J. Chem. Phys.* **2017**, *144*, 214302.
- [30] a) J. Gierschner, J. Shi, B. Milián-Medina, D. Roca-Sanjuán, S. Varghese, S. Y. Park, *Adv. Opt. Mater.* **2021**, *9*, 2002251; b) A. Liess, A. Lv, A. Arjona-Esteban, D. Bialas, A.-M. Krause, V. Stepanenko, M. Stolte, F. Würthner, *Nano Lett.* **2017**, *17*, 1719.
- [31] a) F. Meinardi, M. Cerminara, A. Sassella, R. Bonifacio, R. Tubino, *Phys. Rev. Lett.* **2003**, *91*, 247401; b) T. Nematiram, D. Padula, A. Troisi, *Chem. Mater.* **2021**, *33*, 3368; c) M. Voigt, A. Langner, P. Schouwink, J. M. Lupton, R. F. Mahrt, M. Sokolowski, *J. Chem. Phys.* **2007**, *127*, 114705; d) S.-H. Lim, T. G. Bjorklund, F. C. Spano, C. J. Bardeen, *Phys. Rev. Lett.* **2004**, *92*, 107402.
- [32] M. S. Bradley, J. R. Tischler, V. Bulović, *Adv. Mater.* **2005**, *17*, 1881.
- [33] a) N. A. Leclaire, M. Li, A. C. Véron, A. Neels, H. Jakob, J. R. Reimers, F. A. Nüesch, *Phys. Chem. Chem. Phys.* **2018**, *20*, 29166; b) H. von Berlepsch, S. Möller, L. Dähne, *J. Phys. Chem. B* **2001**, *105*, 5689; c) O. Koch, E. Lork, M. Kleemeier, S. Wiegand, W. Schröer, Z. Kristallogr. New Cryst. Struct. **1999**, *214*, 567; d) J. Delaney, M. Morrow, C. J. Eckhardt, *Chem. Phys. Lett.* **1985**, *122*, 347; e) J. Tanaka, M. Tanaka, M. Hayakawa Bull, *Chem. Soc. Jpn.* **1980**, *53*, 3109; f) K. Nakatsu, H. Yoshioka, H. Morishita, *Acta Crystallogr. Sect. B* **1977**, *33*, 2181; g) A. P. Marchetti, C. D. Salzberg, E. I. P. Walker, *J. Chem. Phys.* **1976**, *64*, 4693; h) B. Dammeier, W. Hoppe, *Acta Crystallogr. Sect. B* **1971**, *27*, 2364.
- [34] J. Knoester, *J. Lumin.* **1992**, *53*, 101.
- [35] J. J. Hopfield, *Phys. Rev.* **1958**, *112*, 1555.
- [36] J. J. Hopfield, D. G. Thomas, *Phys. Rev.* **1963**, *132*, 563.
- [37] a) J. Knoester, *Phys. Rev. Lett.* **1992**, *68*, 654; b) M. Bamba, H. Ishihara, *Phys. Stat. Sol. C* **2009**, *6*, 128; c) M. Bamba, H. Ishihara, *Phys. Rev. B* **2009**, *80*, 125319; d) M. Bamba, H. Ishihara, *J. Phys. Soc. Jpn.* **2009**, *78*, 043701.
- [38] a) A. S. Davydov, *Sov. Phys. Usp.* **1964**, *7*, 145; b) A. S. Davydov, *Theory of Molecular Excitons*, Plenum, New York, **1971**.
- [39] J. Borwein, M. Glasser, R. McPhedran, J. Wan, I. Zucker, *Lattice Sums Then and Now*, Cambridge University Press, Cambridge, **2013**.
- [40] a) R. Silbey, J. Jortner, S. A. Rice, *J. Chem. Phys.* **1965**, *42*, 1515; b) M. R. Philpott, J. W. Lee, *J. Chem. Phys.* **1973**, *58*, 595.
- [41] a) I. Brevik, *Phys. Rev. A* **2018**, *98*, 043847; b) S. M. Barnett, R. Loudon, *J. Phys. B* **2006**, *39*, S671; c) S. M. Barnett, *Phys. Rev. Lett.* **2010**, *104*, 070401.
- [42] a) C. F. Klingshirn, *Semiconductor Optics*, 4th ed., Springer, Berlin, **2012**; b) K. Henneberger, *Phys. Rev. Lett.* **1998**, *80*, 2889.
- [43] P. G. de Gennes, *The Physics of liquid crystals*, Clarendon Press, Oxford, **1974**.
- [44] G. Lakhwani, S. C. J. Meskers, *J. Phys. Chem. Lett.* **2011**, *2*, 1497.
- [45] R. M. A. Azzam, N. M. Bashara, *Ellipsometry and polarized light*, North-Holland, Amsterdam, **1987**.
- [46] a) O. L. Polyansky, A. G. Császár, S. V. Shirin, N. F. Zobov, P. Barletta, J. Tennyson, *Science* **2003**, *299*, 539; b) P. Pytko, K. G. Dyall, A. G. Császár, G. Tarczay, O. L. Polyansky, J. Tennyson, *Phys. Rev. A* **2001**, *63*, 024502.
- [47] A. Salam, *WIREs Comput. Mol. Sci.* **2015**, *5*, 178.
- [48] E. Palikot, M. Stanke, L. Adamowicz, *J. Phys. B* **2020**, *53*, 225101.
- [49] J. G. Balcerzak, M. Lesiuk, R. Moszynski, *Phys. Rev. A* **2017**, *96*, 052510.
- [50] Z. Ao, S. Ashhab, F. Yoshihara, T. Fuse, K. Kakuyanagi, S. Saito, T. Aoki, K. Semba, *Sci. Rep.* **2023**, *13*, 11340.
- [51] J. Linderberg, *Int. J. Quantum Chem.* **1967**, *1*, 719.
- [52] A. F. Silva, L. J. Duarte, P. L. A. Popelier, *Struct. Chem.* **2020**, *31*, 507.
- [53] a) S. Ehrlich, J. Moellmann, S. Grimme, *Acc. Chem. Res.* **2013**, *46*, 916; b) E. Caldeweyher, S. Ehler, A. Hansen, H. Neugebauer, S. Spicher, C. Bannwarth, S. Grimme, *J. Chem. Phys.* **2019**, *150*, 154122.
- [54] a) S. C. J. Meskers, G. Lakhwani, *J. Chem. Phys.* **2016**, *145*, 194703; b) S. C. J. Meskers, *J. Chem. Phys.* **2018**, *148*, 114703.
- [55] J. Keeling, S. Kéna-Cohen, *Annu. Rev. Phys. Chem.* **2020**, *71*, 435.
- [56] K. E. McGhee, R. Jayaprakash, K. Georgiou, S. L. Burg, D. G. Lidzey, *J. Mater. Chem. C* **2022**, *10*, 4187.
- [57] R. Pandya, R. Y. S. Chen, Q. Gu, J. Sung, C. Schnedermann, O. S. Ojambati, R. Chikkaraddy, J. Gorman, G. Jacucci, O. D. Onelli, T. Willhammar, D. N. Johnstone, S. M. Collins, P. A. Midgley, F. Auras, T. Baikie, R. Jayaprakash, F. Mathevet, R. Soucek, M. Du, A. M. Alvertis, A. Ashoka, S. Vignolini, D. G. Lidzey, J. J. Baumberg, R. H. Friend, T. Barisien, L. Legrand, A. W. Chin, J. Yuen-Zhou, S. K. Saikin, P. Kukura, A. J. Musser, A. Rao, *Nat. Commun.* **2021**, *12*, 6519.
- [58] J. Tang, J. Zhang, Y. Lv, H. Wang, F. F. Xu, C. Zhang, L. Sun, J. Yao, Y. S. Zhao, *Nat. Commun.* **2021**, *12*, 3265.
- [59] F. Herrera, F. C. Spano, *Phys. Rev. A* **2017**, *95*, 053867.
- [60] J. J. M. Halls, C. A. Walsh, N. C. Greenham, E. A. Marseglia, R. H. Friend, S. C. Moratti, A. B. Holmes, *Nature* **1995**, *376*, 498.
- [61] G. Yu, J. Gao, J. C. Hummelen, F. Wudl, A. J. Heeger, *Science* **1995**, *270*, 1789.
- [62] G. Zhang, J. Zhao, P. C. Y. Chow, K. Jiang, J. Zhang, Z. Zhu, J. Zhang, F. Huang, H. Yan, *Chem. Rev.* **2018**, *118*, 3447.
- [63] C. Yan, S. Barlow, Z. Wang, H. Yan, A. K.-Y. Jen, S. R. Marder, X. Zhan, *Nat. Rev. Mater.* **2018**, *3*, 18003.
- [64] a) J. Schmidt, in *Physics and Chemistry of Materials with Low-Dimensional Structures*, Vol. 2. (Ed. J. Fünfschilling) Springer, Dordrecht, **1989**, pp. 51–111; b) J. Grüne, G. Lodi, A. J. Gillett, B. Stähly, S. Lulei, M. Kotova, Y. Olivier, V. Dyakonov, A. Sperlich, *Adv. Funct. Mater.* **2023**, *33*, 2212640; c) M. L. Williams, I. Schlesinger, R. M. Jacobberger, M. R. Wasielewski, *J. Am. Chem. Soc.* **2022**, *144*, 18607; d) M. Hariharan, G. D. Scholes, *J. Phys. Chem. Lett.* **2022**, *13*, 365.

- [65] a) A. J. Gillett, A. Privitera, R. Dilmurat, A. Karki, D. Qian, A. Pershin, G. Londi, W. K. Myers, J. Lee, J. Yuan, S.-J. Ko, M. K. Riede, F. Gao, G. C. Bazan, A. Rao, T.-Q. Nguyen, D. Beljonne, R. H. Friend, *Nature* **2021**, 597, 666; b) A. Montanaro, K. H. Park, F. Fassioli, F. Giusti, D. Fausti, G. D. Scholes, *J. Phys. Chem. C* **2023**, 127, 13712.
- [66] T. M. Clarke, J. R. Durrant, *Chem. Rev.* **2010**, 110, 6736.
- [67] S. N. Hood, I. Kassal, *J. Phys. Chem. Lett.* **2016**, 7, 4495.
- [68] S. Athanasopoulos, H. Bässler, A. Köhler, *J. Phys. Chem. Lett.* **2019**, 10, 7107.
- [69] T. Offermans, S. C. J. Meskers, R. A. J. Janssen, *Chem. Phys.* **2005**, 308, 125.
- [70] V. C. Nikolis, A. Mischok, B. Siegmund, J. Kublitski, X. Jia, J. Benduhn, U. Hörmann, D. Neher, M. C. Gather, D. Spoltore, K. Vandewal, *Nat. Commun.* **2019**, 10, 3706.
- [71] a) D. B. Riley, P. Meredith, A. Armin, O. J. Sandberg, *J. Phys. Chem. Lett.* **2022**, 13, 4402; b) M. T. Sajjad, A. Ruseckas, L. K. Jagadamma, Y. Zhang, I. D. W. Samuel, *J. Mater. Chem. A* **2020**, 8, 15687; c) A. J. Sneyd, T. Fukui, D. Paleček, S. Prodhan, I. Wagner, Y. Zhang, J. Sung, S. M. Collins, T. J. A. Slater, Z. Andaji-Garmaroudi, L. R. MacFarlane, J. D. Garcia-Hernandez, L. Wang, G. R. Whittell, J. M. Hodgkiss, K. Chen, D. Beljonne, I. Manners, R. H. Friend, A. Rao, *Sci. Adv.* **2021**, 7, 32.
- [72] a) I. Song, J. Ahn, H. Ahn, S. H. Lee, J. Mei, N. A. Kotov, J. H. Oh, *Nature* **2023**, 617, 92; b) L. Wan, R. Zhang, E. Cho, H. Li, V. Coropceanu, J.-L. Brédas, F. Gao, *Nat. Photonics* **2023**, 17, 649; c) D. Zhu, W. Jiang, Z. Ma, J. Feng, X. Zhan, C. Lu, J. Liu, J. Liu, Y. Hu, D. Wang, Y. S. Zhao, J. Wang, Z. Wang, L. Jiang, *Nat. Commun.* **2022**, 13, 3454; d) M. D. Ward, W. Shi, N. Gasparini, J. Nelson, J. Wade, M. J. Fuchter, *J. Mater. Chem. C* **2022**, 10, 10452; e) L. Liu, Z. Wei, S. C. J. Meskers, *Adv. Mater.* **2023**, 35, 2209730.

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